

① 강의노트 개념 먼저 훑기

② 추천 순서대로 10문제 (목표 2시간)

③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- LMS(=MMSE) = $E[\Theta|X]$ · 조건부 MSE = $\text{var}(\Theta|X)$ (Law of Total Variance)
- MAP = argmax 사후 · ML = argmax 우도 · uniform prior $\Rightarrow$ MAP = ML
- 베이즈 4버전 ($\Theta \cdot X$ 이산/연속) · evidence $f_X(x) = \int$ 또는 $\sum$
- 컬레: Beta-Binomial · jointly Gaussian $\Rightarrow$ MMSE 선형 (MAP = LMS)
- 명명분포: Exp · Poisson · Gauss · Erlang/Gamma · Beta · Binomial

문항 목차 (총 10문항)

- [1] L11 · PSP 12.1.2 — MMSE · blind estimate ●
- [2] L11 · PSP 12.1.1 — MMSE · 무기역성 ●
- [3] L11 · PSP 12.1.5 — MMSE · 조건부 MSE ■
- [4] L11 · PSP 12.1.6 — Gaussian MMSE ■
- [5] L11 · PSP 12.3.1 — MAP · ML ■
- [6] L11 · PSP 12.3.2 — MAP · ML · MMSE (Poisson) ◆
- [7] L11 · PSP 12.3.3 — Bayes · MMSE/MAP/ML ◆
- [8] L11 · PSP 12.3.4 — Beta-Binomial 컬레 ◆
- [9] L11 · PSP 12.1.7 — MMSE · 변환 ◆
- [10] L11 · PSP 12.1.8 — Bayes · 우도비 ◆

Problem 1 (난이도 ●)

12.1.2 ● X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 6(y-x) & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_X(x)$?
- (b) What is the blind estimate $\hat{x}_B$?
- (c) What is the minimum mean square error estimate of X given $X < 0.5$?
- (d) What is $f_Y(y)$?
- (e) What is the blind estimate $\hat{y}_B$?
- (f) What is the minimum mean square error estimate of Y given $Y > 0.5$?

Problem 2 (난이도 ●)

12.1.1● Generalizing the solution of Example 12.2, let the call duration T be an exponential (λ) random variable. For $t_0 > 0$, show that the minimum mean square error estimate of T , given that $T > t_0$ is

$$\hat{T} = t_0 + \mathbf{E}[T].$$

Problem 3 (난이도 ■)

12.1.5■ X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_{X|Y}(x|y)$?
- (b) What is $\hat{x}_M(y)$, the MMSE estimate of X given $Y = y$?
- (c) What is

$$e^*(0.5) = \text{E} \left[(X - \hat{x}_M(0.5))^2 | Y = 0.5 \right],$$

the minimum mean square error of the estimate of X given $Y = 0.5$?

Problem 4 (난이도 ■)

12.1.6■ A signal X and noise Z are independent Gaussian $(0, 1)$ random variables, and $Y = X + Z$ is a noisy observation of the signal X . Usually, we want to use Y to estimate of X ; however, in this problem we will use Y to estimate the noise Z .

- (a) Find $\hat{Z}(Y)$, the MMSE estimator of Z given Y .
- (b) Find the mean squared error $e = E[(Z - \hat{Z}(Y))^2]$.

Problem 5 (난이도 ■)

12.3.1 ■ Suppose that in Quiz 12.3, R , measured in meters, has a uniform PDF over $[0, 1000]$. Find the MAP estimate of R given $X = x$. In this case, are the MAP and ML estimators the same?

Problem 6 (난이도 ◆)

12.3.2◆ Let R be an exponential random variable with expected value $1/\mu$. If $R = r$, then over an interval of length T , the number of phone calls N that arrive at a telephone switch has a Poisson PMF with expected value rT .

- (a) Find the MMSE estimate of N given R .
- (b) Find the MAP estimate of N given R .
- (c) Find the ML estimate of N given R .

Problem 7 (난이도 ♦)

12.3.3♦ Let R be an exponential random variable with expected value $1/\mu$. If $R = r$, then over an interval of length T , the number of phone calls N that arrive at a telephone switch has a Poisson PMF with expected value rT .

- (a) Find the MMSE estimate of R given N .
- (b) Find the MAP estimate of R given N .
- (c) Find the ML estimate of R given N .

Problem 8 (난이도 ♦)

12.3.4♦ Flip a coin n times. For each flip, the probability of heads is $Q = q$ independent of all other flips. Q is a uniform $(0, 1)$ random variable. K is the number of heads in n flips.

- (a) What is the ML estimator of Q given K ?
- (b) What is the PMF of K ? What is $E[K]$?
- (c) What is the conditional PDF $f_{Q|K}(q|k)$?
- (d) Find the MMSE estimator of Q given $K = k$.

Problem 9 (난이도 ♦)

12.1.7♦ Random variable $Y = X - Z$ is a noisy observation of the continuous random variable X . The noise Z has zero expected value and unit variance and is independent of X . Find the conditional expectation $E[X|Y]$.

Problem 10 (난이도 ♦)

12.1.8♦ In a BPSK communication system, a source wishes to communicate a random bit X to a receiver. The possible inputs $X = 1$ and $X = -1$ are equally likely. In this system, the source transmits X multiple times. In the i th transmission, the receiver observes $Y_i = X + W_i$. After n transmissions of X , the receiver has observed $\mathbf{Y} = \mathbf{y} = [y_1 \ \cdots \ y_n]'$.

- (a) Find $\hat{X}_n(\mathbf{y})$, the MMSE estimate of X given the observation $\mathbf{Y} = \mathbf{y}$. Express your answer in terms of the likelihood ratio

$$L(\mathbf{y}) = \frac{f_{\mathbf{Y}|X}(\mathbf{y}|-1)}{f_{\mathbf{Y}|X}(\mathbf{y}|1)}.$$

- (b) Simplify your answer when the W_i are iid Gaussian $(0, 1)$ random variables, independent of X .